

WORKS MANAGEMENT & COST ANALYSIS

Underground CBRN-Hardened Ops Room (22.0m × 6.2m)

Complete BOQ estimation, concrete and rebar takeoff, labour analysis, and total project capital costing aligned with Master Construction Schedule R0.



Duration: 224 Working Days Best Rating: 383 kPa M35
Concrete & Fe500D Rebar

Works Management & WBS Overview

Master Schedule Breakdown

The construction spans **224 working days**, starting with site establishment on 02-11-2026 and culminating in comprehensive integration testing and handover on 26-07-2027.

- **Initial & Site Prep:** 4 days (Clearing, water/electricity setup)
- **Substructure Works:** 40 days (Deep mass excavation & raft)
- **Superstructure & RCC:** 82 days (Walls, 900mm roof slab)
- **Finishing & MEP/CBRN:** 112 days (Pinning, blast valves, fitouts)

Key Milestones & Constraints

A rigid sequencing protocol governs structural hardening, curing times, and radiological protection:

- **Milestone 1:** Mat Raft & Periphery Starter casting to -3.2m RL.
- **Milestone 2:** Monolithic 900mm roof pressure slab casting & 28-day cure.
- **Milestone 3:** 2.0m engineered cover placement, CBRN leak tests & commissioning.
- **Working Framework:** 6 working days/week, strict QMS checking.

Concrete Volume Takeoff (M35 / M15)

Structural Element	Grade	Dimensions / Specification	Volume (m³)
Bottom Blinding PCC	M15	100 mm under mat slab (23.2m × 7.4m)	17.17
Monolithic Mat Raft	M35	600 mm thick continuous slab (22.0m × 6.2m)	81.84
Perimeter Shear Walls (W1-W4)	M35	600 mm thick, 3.2m clear height	108.29
Internal Blast & Dividing Walls (W5-W7)	M35	200 mm gas-tight & 400 mm shaft walls	24.32
Pressure Roof Slab	M35	900 mm thick slab with 500 mm corner haunches	114.72
Entry Headhouse & Approach Stairwell	M35	400 mm walls, 500 mm roof & approach channel	46.76
Burster Slab (in 2.0m Earth Cover)	M35	300 mm sacrificial penetrator disruption slab	57.60
Sentry Post & Sump Internal Structures	M30/M35	Plinths, columns, two-way slab & Bay 5 sump pit	19.80
Total Project Concrete Quantity	470.50 m³		

Steel Reinforcement Estimate (Fe500D)

Bar Dia (Φ)	Application / Structural Location	Net Weight (kg)	Wastage (5%)	Total (Tonnes)
8 mm	Sentry slab distribution, column ties & spacers	1,120.00	56.00	1.18 T
10 mm	Internal staircase distribution & trimmer rebars	1,480.00	74.00	1.55 T
12 mm	Shear wall links, burster slab mesh, Bay 5/6 wall	9,390.00	469.50	9.86 T
16 mm	600mm shear walls & mat foundation double curtains	24,140.00	1,207.00	25.35 T
20 mm	Headhouse roof, shaft walls (W6/W7), corner haunches	8,740.00	437.00	9.18 T
25 mm	900mm roof pressure slab main flexural steel (T25@150)	19,690.00	984.50	20.67 T
Total Rebar Requirement	64,560.00 kg	3,228.00 kg	67.79 Tonnes	

BOQ: Survey, Earthwork & Substructure

Item Description	Unit	Qty	Material (₹)	Labour (₹)	Total Cost (₹)
Geotechnical confirmatory boreholes & core tests	No	4	2,00,000	34,660	2,34,660
Site clearance, surface grubbing & leveling	Sqm	1,800	80,640	15,860	96,500
Mass excavation in overburden & Deccan basalt (6.8m)	Cum	1,539	13,08,150	2,04,420	15,12,570
Formwork for mat foundation & Bay 5 sump pit	Sqm	125	51,250	31,500	82,750
PCC 1:3:6 (M15) 100mm foundation blinding	Cum	17.17	94,435	11,396	1,05,831
Monolithic mat raft RCC casting (M35)	Cum	81.84	7,07,098	54,984	7,62,082
Pre-applied HDPE membrane + screed & waterstops	Sqm	280	3,10,200	22,608	3,32,808
Subtotal Parts I & II	₹	31,96,197			

BOQ: Superstructure Hardened Concrete

Item Description	Unit	Qty	Material (₹)	Labour (₹)	Total Cost (₹)
Formwork & staging for 600mm perimeter walls	Sqm	480	2,78,400	1,33,488	4,11,888
Perimeter (600mm) & blast dividing (400mm) walls M35	Cum	132.61	11,45,750	91,168	12,36,918
Heavy falsework staging for 900mm roof slab	Sqm	145	94,250	1,22,400	2,16,650
Pressure roof slab (900mm) casting monolithic M35	Cum	114.72	9,91,181	82,476	10,73,657
Entry Headhouse walls (400mm) & roof (500mm)	Cum	28.56	2,46,758	33,372	2,80,130
Approach stairwell box & internal dog-leg staircase	Cum	23.00	2,01,408	33,372	2,34,780
Engineered burster slab (300mm) & Sentry Post frame	Cum	76.20	6,30,420	67,554	6,97,974
Subtotal Part III (Superstructure Hardened RCC)	₹	42,61,645			

BOQ: Rebar, Protective Systems & Ancillaries

Item Description	Unit	Qty	Material (₹)	Labour (₹)	Total Cost (₹)
High-yield Fe500D rebar (Roof, walls, mat, headhouse)	Tonne	67.79	52,87,620	6,26,652	59,14,272
EMP Faraday cage continuity welding & perimeter earth	Item	1	3,80,000	38,880	4,18,880
Blast Doors BD-1 & BD-2 (≥7 bar reflected, gas-tight)	Sets	2	25,00,000	50,400	25,50,400
Emergency blast escape hatches (900mm clear dia)	Sets	2	9,00,000	22,248	9,22,248
Blast valves BV-1 to BV-5 (DN100 and DN350 trains)	Sets	5	11,20,000	20,736	11,40,736
CBRN filtration package (2 × 300 m³/h N+1 trains)	Sets	2	37,00,000	72,576	37,72,576
EMP Zone-2 shielded enclosure (3.0 × 3.0 × 2.4m)	Item	1	8,50,000	44,496	8,94,496
Berm backfilling, drainage pumps, plaster & finishes	Misc	-	11,09,600	1,94,472	13,04,072
Subtotal Parts IV & V	₹		1,68,78,180		

Total Project Cost Analysis

₹ 2.43 Cr

BASIC COST

470.5 m³

TOTAL CONCRETE

67.79 T

TOTAL STEEL

224 Days

TOTAL DURATION

Cost Head Description	Basis of Increase	Amount Incurred (₹)
Total Basic Construction Cost (Parts I - V)	Direct material + labour subtotal	2,43,36,022.00
Physical & Design Contingencies	3.0% on Basic Cost	7,30,081.00
Site Water & Electricity Provisions	1.0% on Basic Cost	2,43,360.00
Specialist Engineering & CBRN/EMP Consultant	6.0% on Basic Cost	14,60,161.00
Site Supervision, QA/QC & Witness Testing	2.0% on Basic Cost	4,86,720.00
Statutory Clearances & Defense Liaisoning	1.0% on Basic Cost	2,43,360.00
Contractor Overhead & Profit Margin	10.0% on Basic Cost	24,33,602.00

FINAL CONSOLIDATED PROJECT ESTIMATE

Includes all civil hardening, specialist MEP, CBRN filtration & testing

₹ 3,00,33,306.00